National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task 13

Task 1: Toggle LED Simulation Using Binary Semaphore

Description:
Create a program where two threads simulate toggling an LED on and off alternately. Use a binary semaphore to control which thread can toggle the LED at a time.
Hint:

- One thread turns the LED "ON", the other turns it "OFF".
- Use `sem_wait` and `sem_post` to control the toggling sequence.
- Initialize the semaphore to allow the "ON" thread to start first.

Task 2: Simple Login Authentication Simulation

Description:
Create a program that simulates a login system with two threads:

- One thread verifies the username.
- Another thread verifies the password.
  Use a binary semaphore to make sure that password verification only happens after username verification.
  Hint:
- Username thread should signal the password thread after successful username verification.

- Use `sem_wait` for the password thread to block until the username thread signals.

Task 3: Producer-Consumer Problem Using Binary Semaphore (25 mins)

Description:
Create a program that simulates a producer and a consumer using binary semaphores.

- The producer thread generates a random number and "produces" it.
- The consumer thread "consumes" the number after it is produced.
- Use a binary semaphore to ensure the consumer waits until the producer produces a number.

Hint:

- Use `sem_wait` for the consumer to block until the producer signals.
- Use `sem_post` for the producer to signal the consumer.
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Task 4: Maximum Value Search with Semaphore Synchronization

Description:
Create a program where multiple threads search different parts of an array for the maximum value.

- Use a binary semaphore to ensure only one thread updates the shared maximum variable at a time.

- The main function should display the maximum value after all threads finish.
  Hint:

- Use `sem_wait` to lock before updating the shared max value.

- `sem_post` after the update

THE END
